robogoomy | Ilia + EXPO-RT

BEHAVIOR-1K 2026: a transparent partial submission

- Leonard Yoon / Independent

- Frozen released actor + task-specific action selection and editing
- 36 development rollouts / 12 separate final rollouts
- All final results retained; no additional research rollouts

Two evaluation panels

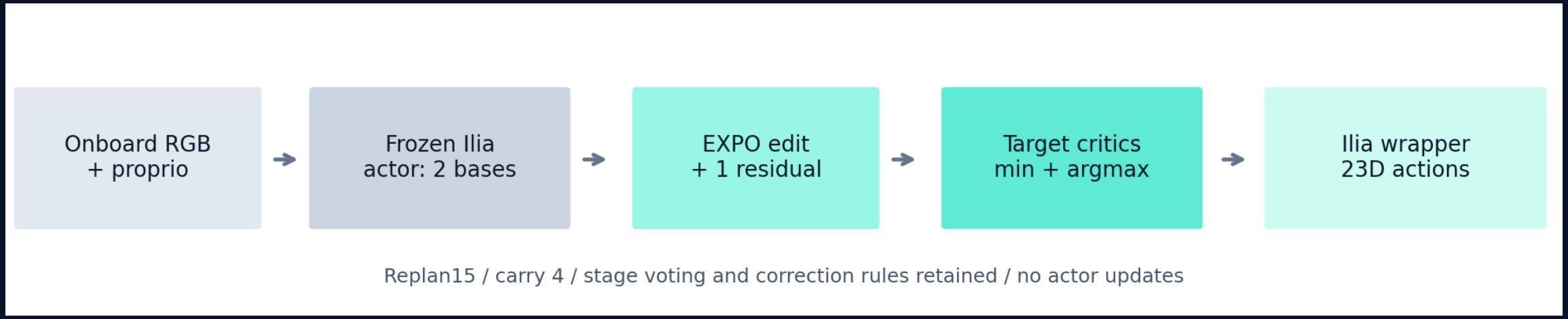
Development comparison and final submission stay separate

- Pilot: 6 tasks x 3 instances x A0/RT = 36 physical rollouts
- Pilot instances: public0 (native301), training23 and24
- Final: tasks002,001,005 x public0-3 x one rollout = 12
- Final panel is RT-only; no matched A0 control

Policy architecture

The original actor is frozen; auxiliary networks adapt

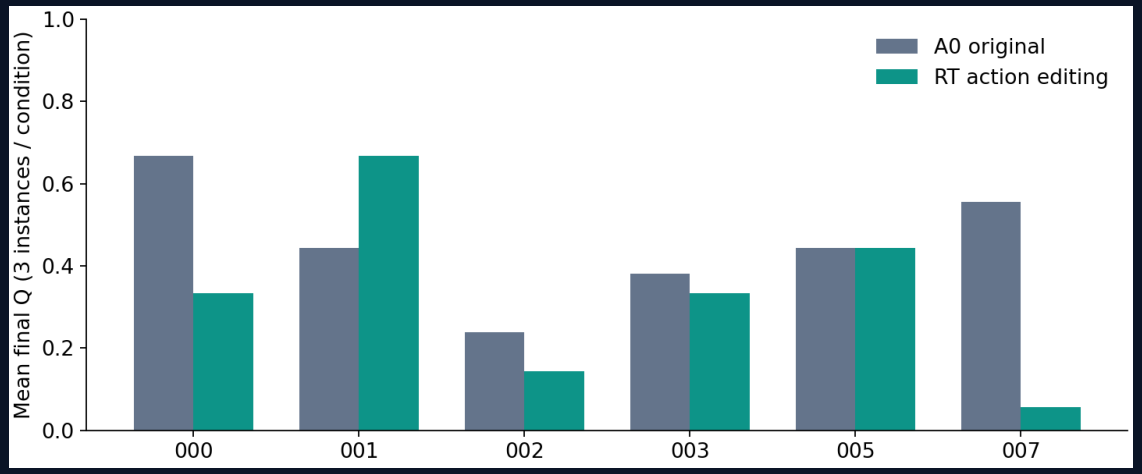
- Native Ilia: VLM, action expert, stage voting, correlated sampling
- RT: two base candidates + one small residual edit
- Target-critic minimum selects a candidate; correction rules follow
- 30-step chunk / execute 15 / carry 4 / edit scale 0.01



Development results

Partial progress gains did not improve aggregate success

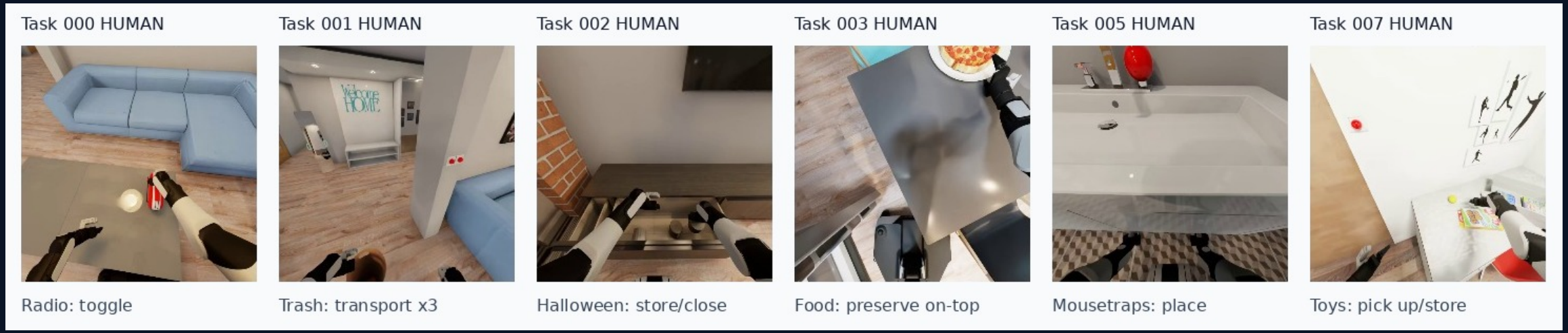
- Mean final Q: A0 0.455 -> RT 0.329
- Full success: 5/18 -> 1/18
- Task001 Q +0.222; task007 Q -0.500
- Small development panel; no general RT-improvement claim



Six household tasks

The development pilot spans navigation and manipulation

- 000 Radio / 001 Trash / 002 Halloween decorations
- 003 Plates and food / 005 Mousetraps / 007 Toys
- Different task actors and auxiliaries; no universal-effect claim



Final 12 results

Observed mean Q: 0.224206 | Full success: 0/12

- All 12 complete; original JSON and MP4 retained unchanged
- 002 mean 0.464286 / 001 mean 0 / 005 mean 0.208333
- Time-limited scope; unevaluated instances count as zero

Task	Public 0	Public 1	Public 2	Public 3
002	0.571	0.714	0.571	0.000
001	0.000	0.000	0.000	0.000
005	0.000	0.500	0.000	0.333

What the scores mean

Goal progress and complete success are different

- Q_final: fraction of final goal conditions satisfied
- Observed 12 mean 0.224206; selected30-instance mean 0.089683
- Full100-task / 1000-instance denominator: Q0.00269048
- Official intermediate Qmax and goal traces were not captured

Failure observations

Video evidence supports hypotheses, not hidden-state proof

- Task001: trash remains held; placement/recovery often repeats
- Task002 public 3: limited movement and occluded wrist views
- Task005: partial progress only in public1 and public 3
- Video timestamps, final metrics, and critic Q must be separated

Runtime and deployment

Policy-only release check on the current L40S passed

- Final evaluation: 4x L40S; policy and simulator share each GPU
- 46 logical CPUs; simulator ceilings changed during execution
- Embedded weights: 5 routes, 35 finite actions/route,seeded reset
- Current L40S peak 7317MiB; real24GB hardware test skipped

Reproducible public artifacts

Original files, exact configuration, pinned identities

- Public GitHub: code, evaluation README, reports, video index
- Public HF: 12 JSON + 12 MP4 + final ZIP + image archive
- Unchanged stock R1Pro / full-resolution RGBD wrapper
- Model and archive hashes preserved; credentials/assets excluded

Submission and follow-up

Finish deployment and portal receipt, then analyze evidence

- GitHub Actions publishes the image using automatic authentication
- Digest-pinned image + anonymous artifact download verification
- Save completed form, acceptance ID, and leaderboard screenshots
- No new rollouts now; future F1+RT remains a research idea